Motors and generators

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

SUCCESS CRITERIA - I can...

1. Trace energy through a motor and a generator.
2. Identify magnetic interactions in rotation or induction.
3. Include less useful energy transfers.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “motor” and “generator” to compare your model or result. Sentence starters are optional.

motor: a device converting electrical energy to mechanical output

generator: a device converting mechanical input to electrical output

efficiency: useful output divided by input

01 Bring back an earlier idea

A consistently drawn magnetic-field map has 3 times as many field lines through the same small perpendicular area at A as at B. What does the denser spacing indicate?

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\eta = P_{useful}/P_{input}$; $E_{input} = E_{useful} + E_{losses}$. Motor: electrical to mechanical; generator: mechanical to electrical. Count losses and any stored-energy change.

Worked reasoning: An idealized motor receives 20 J electrically and delivers 12 J mechanically; 8 J goes to other transfers. A generator runs the energy conversion in the opposite useful direction.

Your turn: Draw separate motor and generator energy-flow diagrams.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

Complete two energy chains: battery → motor → turning axle, and turning axle → generator → lamp.
Label each input and useful output.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

04 Build the solution

A motor receives 100 J electrical energy and outputs 60 J useful mechanical energy.
Find efficiency and energy in other forms.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

Motors and generators

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A motor drives a generator that feeds the motor. **Explain** why real losses prevent the loop from running forever without external input.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

An electric field of 6 N/C points left. **Draw** the force direction on +4 C and on -4 C, then compare sizes.

Word help: electric field = force per positive test charge at a location

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

Which device converts mechanical input into electrical output?

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A motor draws the same electrical power in two designs, but one delivers more mechanical power. **Compare** efficiencies and name a measurement needed.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3